

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination - 2025

Course: B.Tech (Common to All)

Semester : II

Subject Code & Name: Programming for Problem Solving (24AF1000ES106)

Max Marks: 60

Date: 24th July 2025

Duration: 3 Hr.

Instructions to the Students:

- 1 Each question carries 12 marks.
- 2 Question No. 1 will be compulsory and include objective-type questions.
- 3 Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- 4 The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- 5 Use of non-programmable scientific calculators is allowed.
- 6 Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Objective type questions. (Compulsory Question)

12

Which of the following is a system software?

- 1 a) Microsoft Word b) Tally c) Turbo C++ d) Operating System 1

Which of the following is used for type casting in C?

- 2 a) #define b) (data_type) c) & d) @ 1

Which of the following is a relational operator in C?

- 3 a) && b) || c) == d) = 1

Which keyword is used to jump to a labeled statement in C?

- 4 a) goto b) jump c) switch d) case 1

Which of the following denotes the address of a variable in C?

- 5 a) * b) && c) & d) @ 1

Which operator is used for bitwise AND operation?

- 6 a) && b) || c) & d) ^ 1

Which of the following loop runs at least once even if the condition is false?

- 7 a) while b) for c) do-while d) switch 1

What will *(ptr + 2) integer array?

- 8 a) Address of third element b) Value of third element 1
c) First element d) Size of array

What is the size of int in C on a 32-bit system?

- 9 a) 8 bytes b) 4 bytes c) 2 bytes d) 1 byte 1

Which type of memory is volatile and loses data when power is turned off?

- 10 a) Hard Disk b) ROM c) RAM d) Pen Drive 1

Which function returns control to the calling function in C?

- 11 a) break b) return c) exit() d) continue 1

What is the output of 10 % 3 in C?

- 12 a) 0 b) 1 c) 3 d) 1 1

Q. 2 Solve the following.	12
A) List and describe the various generations of computers.	6
B) Explain the role of input and output devices in a computer system with suitable examples.	6
Q.3 Solve the following.	12
A) Explain operator precedence and associativity in C with an example.	6
B) Define C tokens. List different types of tokens used in C.	6
Q. 4 Solve Any Two of the following.	12
A) Explain the difference between break, continue, and goto statements with examples.	6
B) List and define and write syntax for different types of loops used in C programming.	6
C) Create a user-defined function to calculate the factorial of a number.	6
Q.5 Solve Any Two of the following.	12
A) Explain the difference between one-dimensional and two-dimensional arrays with syntax and examples.	6
B) Write a C program to find the average of 10 numbers using arrays.	6
C) Compare the use of arrays and pointers for data manipulation in terms of efficiency.	6
Q. 6 Solve Any Two of the following.	12
A) What are structures in C? List the steps to declare and initialize a structure with syntax.	6
B) Explain with syntax different file handling modes in C.	6
C) Write a C program to input and display details of a student using structures.	6
*** End ***	